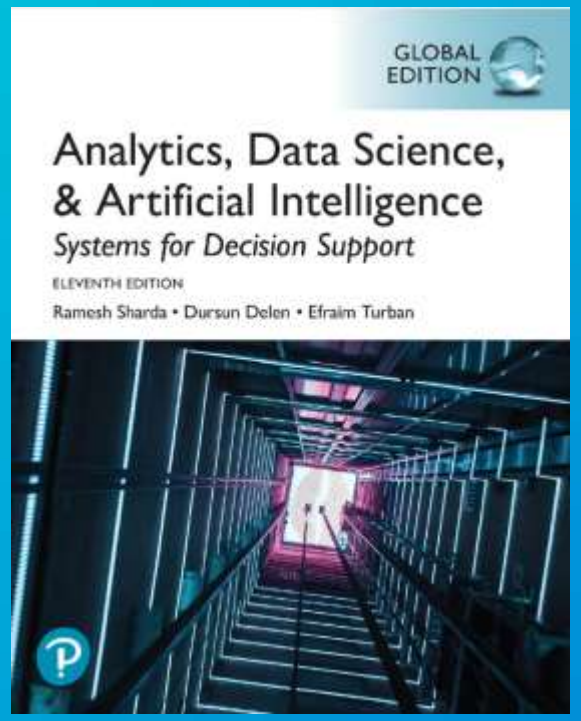
MANAGEMENT SUPPORT SYSTEMS
(415 MIS-3)
CHAPTER - 6



Chapter 7: Text Mining, Sentiment Analysis, and Social Analytics

Analytics, Data Science, & Artificial Intelligence Systems For Decision Support,11ed, Global Edition

Contents

- **7.2** - Text Analytics and Text Mining Overview
- **7.3** - Natural Language Processing (NLP)
- **7.4** - Text Mining Applications
- **7.5** - Text Mining Process
- **7.6** - Sentiment Analysis
- **7.7** - Web Mining Overview
- **7.8** - Search Engines
- **7.9** - Web Usage Mining (Web Analytics)
- **7.10** - Social Analytics

7.2 Text Analytics and Text Mining Overview

- Text Mining Concepts
- Text analytics versus Text Mining
- Data Mining versus Text Mining
- Text Mining Application Area
- Text Mining Terminology

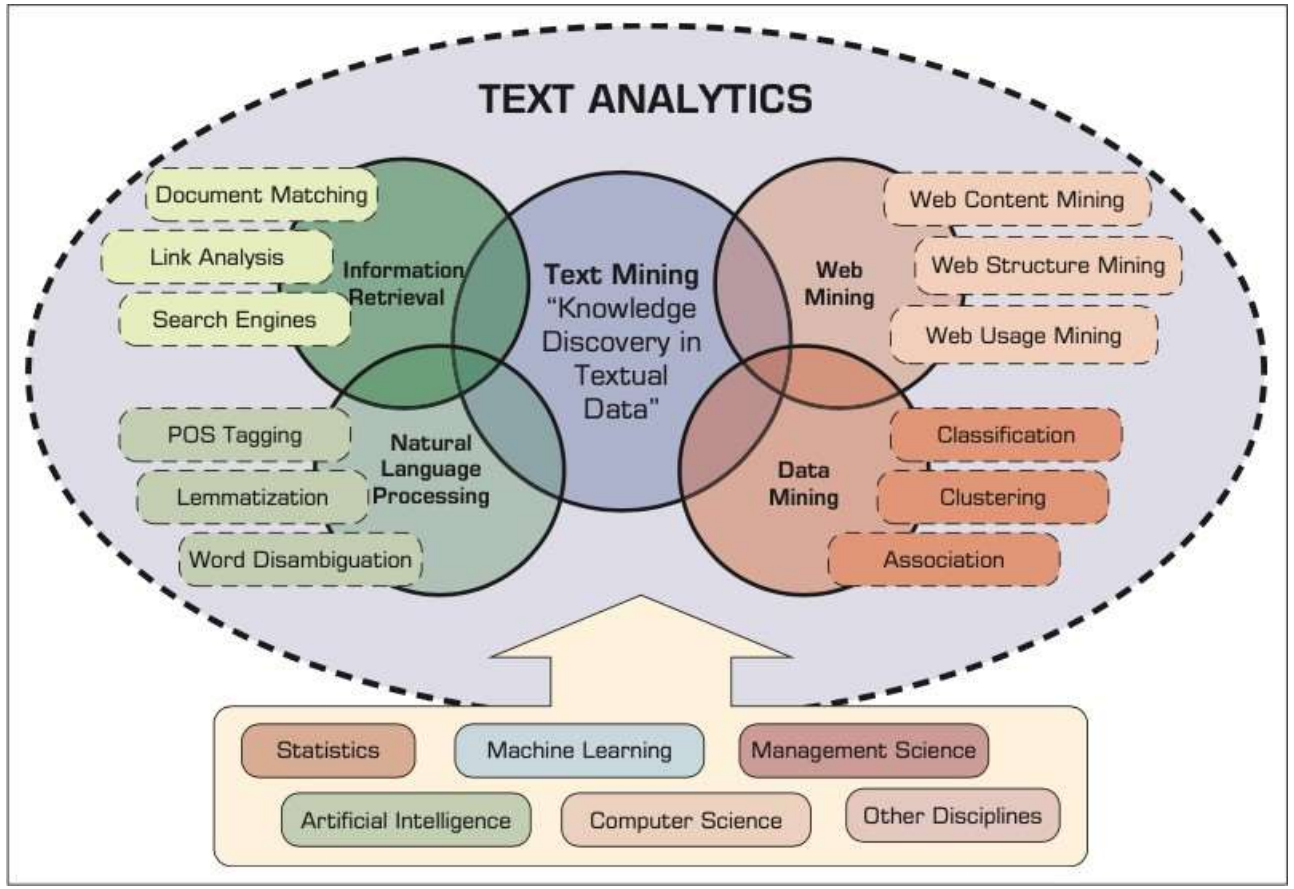
Text Mining Concepts

- Most of corporate data are captured and stored in some sort of unstructured form (e.g., text)
- knowledge is derived from data and information → power in today's business world.
- Businesses that effectively and efficiently tap into their text data sources will have the necessary knowledge to make better decision.
- To stay competitive text mining is needed!
  - E.g., Amadori Group Converts Consumer Sentiments into Near-Real-Time Sales using text mining.

What is Text Mining?

- Text mining is the discovery of knowledge from largely unstructured (text-based) data sources.
- One of the fastest-growing branches of business intelligence field.
- A semiautomated process of extracting patterns (useful information and knowledge) unstructured data sources.

Text Analytics and Text Mining



A relationships between text analytics and text mining along with other related application areas and enabling disciplines

Text analytics versus Text Mining

- Both seek to turn unstructured textual data into actionable information through the application of natural language processing (NLP) and analytic.
- Text analytics is a broader concept that includes: information retrieval (e.g., searching and identifying relevant documents for a given set of key terms), as well as information extraction, data mining, and Web mining.
- Text mining is primarily focused on discovering new and useful knowledge from the textual data sources.
- Even though the two can be defined somewhat differently at times, text analytics and text mining are usually used synonymously.

Data Mining versus Text Mining

- Both has the same purpose of identifying valid, novel, potentially useful, and ultimately understandable patterns in data.
- Both uses the same processes (semi-automate).
- They Differ in input to the process → the nature of the data.

Data mining	Text mining
Input → Structured data found in databases	Input → Unstructured data such as as Word documents, PDF files, text excerpts, and XML files.

- Text mining uses NLP to first induce structure into the text collection and then uses data mining algorithms such as classification, clustering, association, and sequence discovery to extract knowledge from it.

Text Mining Concepts

- Benefits of text mining are obvious
  - E.g., the free-form text-based interactions with customers in the form of complaints (or compliments) can be used to objectively identify product and service characteristics that are deemed to be less than perfect and can be used as input to better product development and service allocations.
- Some text-rich data environments
  - E.g., law (court orders), academic research (research articles), finance (quarterly reports), medicine (discharge summaries), biology (molecular interactions), technology (patent files), and marketing (customer comments).
- Area where the automated processing of unstructured text has had much impact is Electronic communication records (e.g., Email)
  - filter junk e-mail
  - automatically prioritize and categorize e-mail
  - generate automatic responses

Text Mining Application Area

Most popular application areas of text mining:

Information extraction	Identifying key phrases and relationships within text.
Topic tracking	Predicting other documents of interest to the user, based on a user profile and documents that a user views.
Summarization	Summarizing a document to save the reader time.
Categorization	Identifying the main themes of a document and then placing the document into a predefined set of categories based on those themes.
Clustering	Grouping similar documents without having a predefined set of categories.
Concept linking	Connecting related documents by identifying their shared concepts
Question answering	Finding the best answer to a given question through knowledge-driven pattern matching.

Text Mining Terminology

- **Unstructured or semi-structured data**
- **Corpus:** a large and structured set of texts prepared for knowledge discovery.
- **Terms:** single word or multiword phrase extracted directly from the corpus
- **Concepts:** features generated from a collection of documents using manual, statistical, rule-based, or hybrid categorization methodology.
- **Stemming:** process of reducing inflected words to their stem.
- **Stop words:** noise words - filtered out prior to or after processing natural language data (e.g., (*a, an, the*)
- **Synonyms and polysemes**
- **Tokenizing:** a categorized block of text in a sentence.

Text Mining Terminology

- **Term dictionary:** a collection of terms specific to a narrow field that can be used to restrict the extracted terms within a corpus.
- **Word frequency:** number of times a word is found in a specific document.
- **Part-of-speech tagging:** process of marking the words to a particular part of speech (nouns, verbs, adjectives, adverbs, etc.)
- **Morphology:** studies the internal structure of words.
- **Term-by-document matrix:** common representation schema of the frequency-based relationship between the terms and documents (occurrence matrix).
- **Singular value decomposition:**
  - Latent semantic indexing - dimensionality reduction method.

7.3 Natural Language Processing (NLP)

Natural Language Processing (NLP)

- Introducing structure to a collection of text-based documents methods
  - bag-of-words an early approach.
  - natural language processing a new approach.

- The goal of NLP is to move beyond semantics-based text mining (bag-of-words) to a true understanding and processing of natural language that considers grammatical and semantic constraints as well as the context.

- **NLP is:**
  An important component of text mining
  a subfield of artificial intelligence and computational linguistics
  studies the problem of "understanding" the natural human language

Natural Language Processing (NLP)

- A major discussion topics in NLP is the definition and scope of computers “understanding”

  - Natural language is context driven and vague
  - True understanding of meaning requires extensive knowledge of a topic

Will computers ever be able to understand natural language the same way and with the same accuracy that humans do?

Longer way to go to really understand natural human language!

Natural Language Processing (NLP)

- Challenges commonly associated with the implementation of NLP:

  - **Part-of-speech tagging** difficulty in marking up terms in a text (context is all you need!)
  - **Text segmentation** single-word boundaries problem (e.g., Chinese language).
  - **Word sense disambiguation** words with more than one meaning.
  - **Syntax ambiguity** grammar for natural languages is ambiguous
  - **Imperfect or irregular input** (e.g., grammatical errors in texts).
  - **Speech acts** sentence structure might not contain enough information to define acts.

To have algorithms that are capable of automatically reading and obtaining knowledge from text -> Dream of AI community!

Natural Language Processing (NLP)

- To automatically identify the concepts and relationships between them → WordNet
  - A laboriously hand-coded database of English words, their definitions, sets of synonyms, and various semantic relations between synonym sets.
  - A major resource for NLP applications
  - very expensive to build and maintain manually → Need automation
- Sentiment Analysis
  - A technique used to detect favorable and unfavorable opinions toward specific products and services - a settled opinion reflective of one's feelings.
  - SentiWordNet - is an extension of WordNet to be used for sentiment identification.

NLP Task Categories

- NLP has successfully been applied to a variety of domains
  - Question answering
    - answering a question posed in natural language (e.g., chatbots)
  - Automatic summarization
    - creation of a shortened version of a textual document
  - Natural language generation & understanding
  - Machine translation (e.g., google translate)
  - Foreign language reading & writing
  - Speech recognition

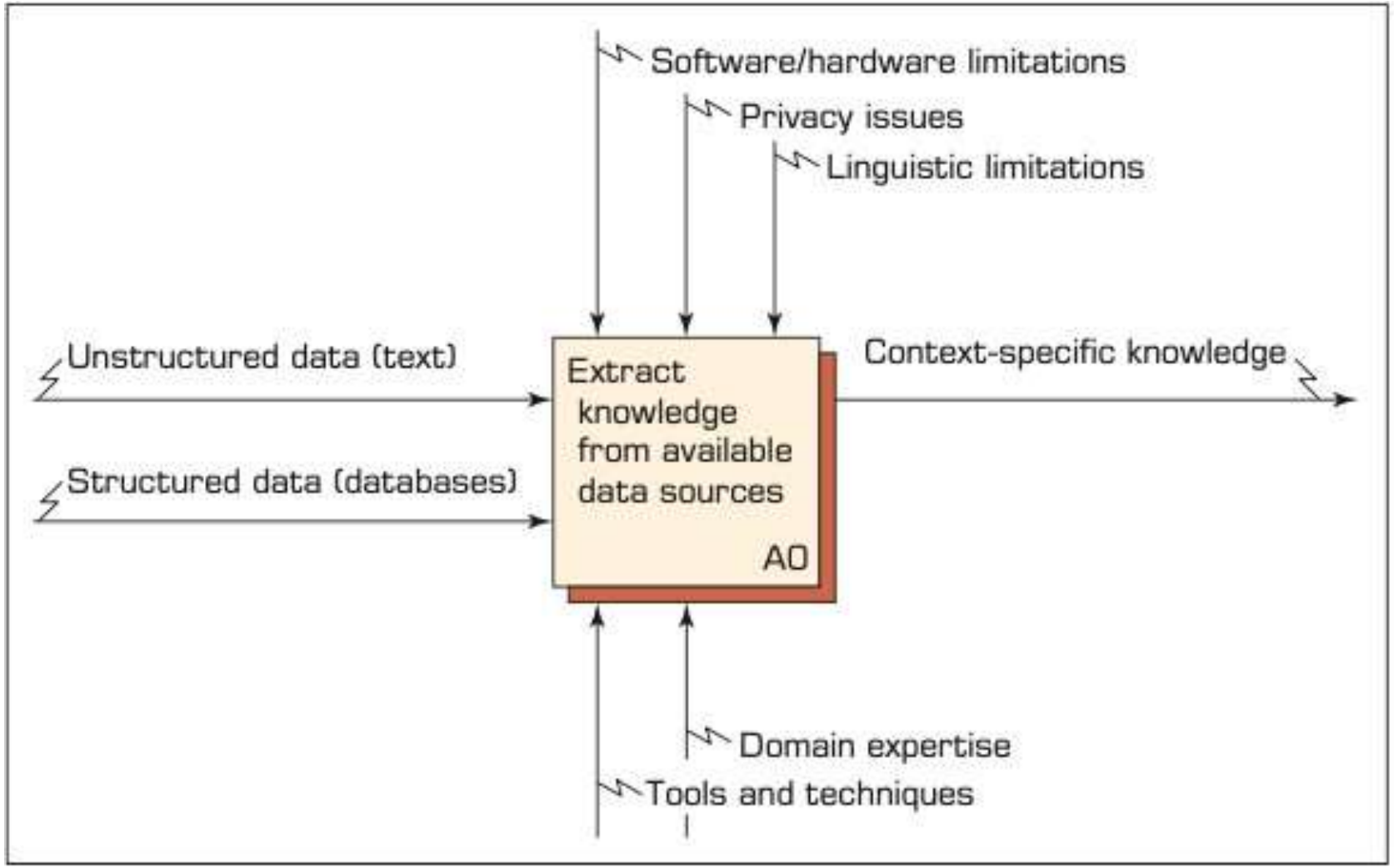
    Conversion of spoken words to machine-readable input
  - Text to speech (a.k.a speech synthesis)
  - Text proofing, optical character recognition

7.4 Text Mining Applications
7.5 Text Mining Process

Text Mining Applications

- **Marketing applications**
  - Increase cross-selling and up-selling by analysing the unstructured data generated by call centres.
  - Enables better CRM
- **Security applications**
  - Deception detection
- **Biomedical Applications**
  - Experimental techniques (e.g., DNA microarray analysis)
- **Academic applications**
  - Research stream analysis

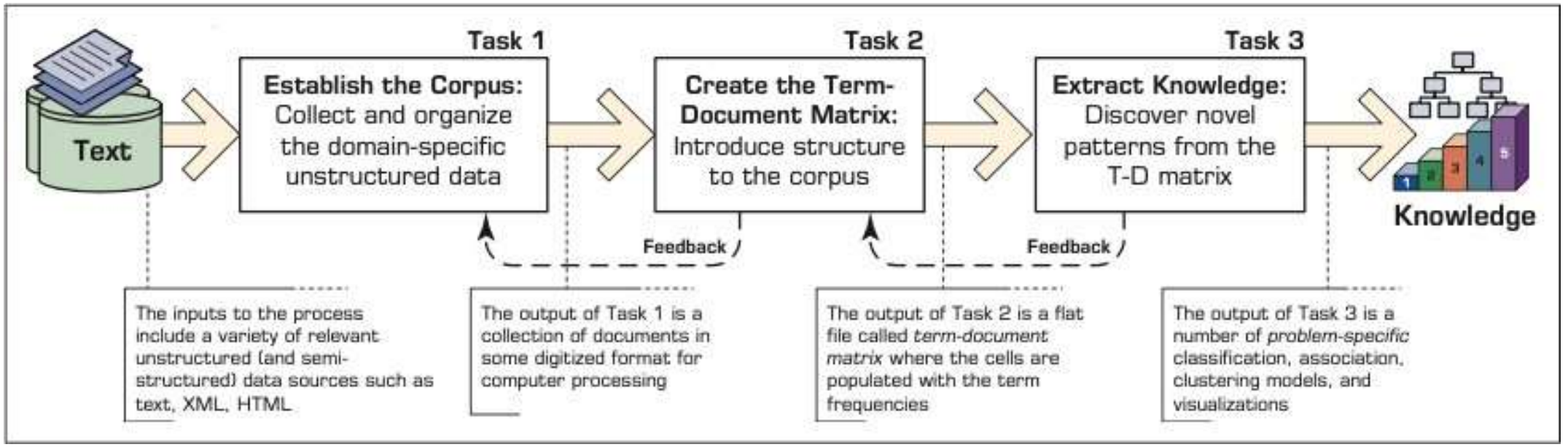
Text Mining Process



Context Diagram for the Text Mining Process

Text Mining Process

- The text mining process can be broken down into three consecutive tasks, each of which has specific inputs to generate certain outputs



The Three-Step/Task Text Mining Process

Text Mining Process

Task 1: Establish the corpus

- Collect all the documents (unstructured data) related to the context (domain of interest) being studied. (e.g., textual documents, XML files, emails, Web pages, and short notes, etc.)
- Transform and organize the collection in a manner such that they are all in the same representational form (e.g., ASCII text files)
- Store the collection in a common place (e.g., in a file folder or a list of links to a collection of Web pages)

Text Mining Process

Task 2: Create the Term-by-Document Matrix (TDM)

- Rows represent the documents and the columns represent the terms.
- The relationships between the terms and documents are characterized by indices (i.e., occurrences of the term)

Terms / Documents	Investment Risk	Project Management	Software Engineering	Development	SAP	...	
Document 1	1			1			
Document 2		1					
Document 3			3		1		
Document 4		1					
Document 5			2	1			
Document 6	1			1			
...							

Text Mining Process

In TDM:

- Are all terms important when characterizing documents?
  - Stop words
  - Predetermined terms (dictionary)
  - Synonyms
  - Stemming

- What is the best representation of the indices (values in cells)?
  - Row counts; binary frequencies; log frequencies;
  - Inverse document frequency

Text Mining Process

Task 2: Create the Term-by-Document Matrix (TDM)

- TDM is a sparse matrix (most of the cells filled with zeros).

Approaches to reduce the dimensionality of the TDM

- Manual - a domain expert goes through it and eliminate terms that do not make much sense
- Eliminate terms with very few occurrences in very few documents
- Transform the matrix using singular value decomposition (SVD) - similar to principle component analysis

Text Mining Process

Task 3: Extract the knowledge/novel patterns

Main categories of knowledge extraction methods

- Classification (text categorization)
- Clustering (text natural groupings)
  - Improve search recall
  - Improve search precision
  - Scatter/gather
  - Query-specific clustering
- Association
- Trend Analysis

7.6 Sentiment Analysis

- Sentiment Analysis Overview
- Sentiment Analysis Applications
- Sentiment Analysis Process
- Methods for Polarity Identification

Sentiment Analysis Overview

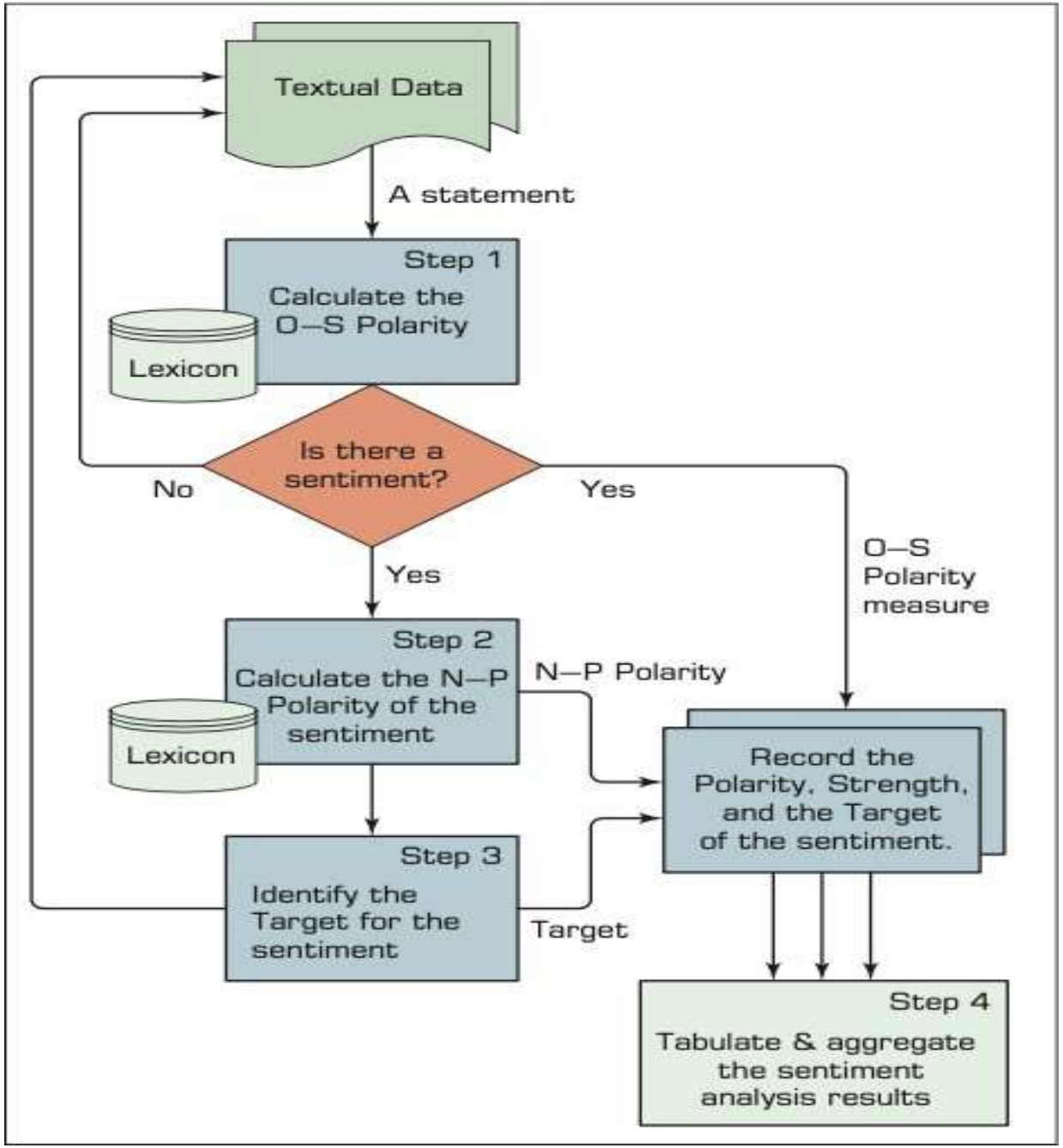
- Sentiment linked to belief, view, opinion, and conviction.
- Sentiment Analysis → opinion mining, subjectivity analysis, appraisal extraction with some connections to affective computing.
- Closely related to computational linguistics, NLP, and text mining.
- Seeks to find out “What do people feel about a certain topic?”
- Deals with differentiating between two classes, positive and negative, versus Neutral - a range of polarity.
- Two types of sentiment: explicit (e.g., “It’s a wonderful day”) or implicit in which text implies an opinion (e.g., “The handle breaks too easily”).

Sentiment Analysis Applications

- Some application examples that illustrate the power and the widespread coverage of sentiment analysis:
  - Voice of the customer (VOC) - an integral part of an analytic CRM
  - Voice of the Market (VOM) - understanding aggregate opinions and trends.
  - Voice of the Employee (VOE)
  - Brand Management
  - Financial Markets
  - Politics
  - Government Intelligence

Sentiment Analysis Process

Multistep Process to Sentiment Analysis



Sentiment Analysis Process

Step 1 – Sentiment Detection

- comes after the retrieval and preparation of the text documents (also called detection of objectivity).
- The goal is to differentiate between a fact and an opinion: objectivity versus subjectivity

Step 2 – N-P Polarity Classification

- Given an opinionated piece of text, the goal is to classify the opinion as falling under one of two opposing sentiment polarities: negative versus positive

Sentiment Analysis Process

Step 3 – Target Identification

- The goal is to accurately identify the target of the expressed sentiment (e.g., a person, a product, an event, etc.)
- Task level of difficulty depends on domain of the analysis.

Step 4 – Collection and Aggregation

- Once the sentiments of all text data points in the document are identified and calculated, they are aggregated and converted to a single sentiment measure (e.g., summing up the polarities).
- From word level to statement, paragraph or document level.

Methods for Polarity Identification

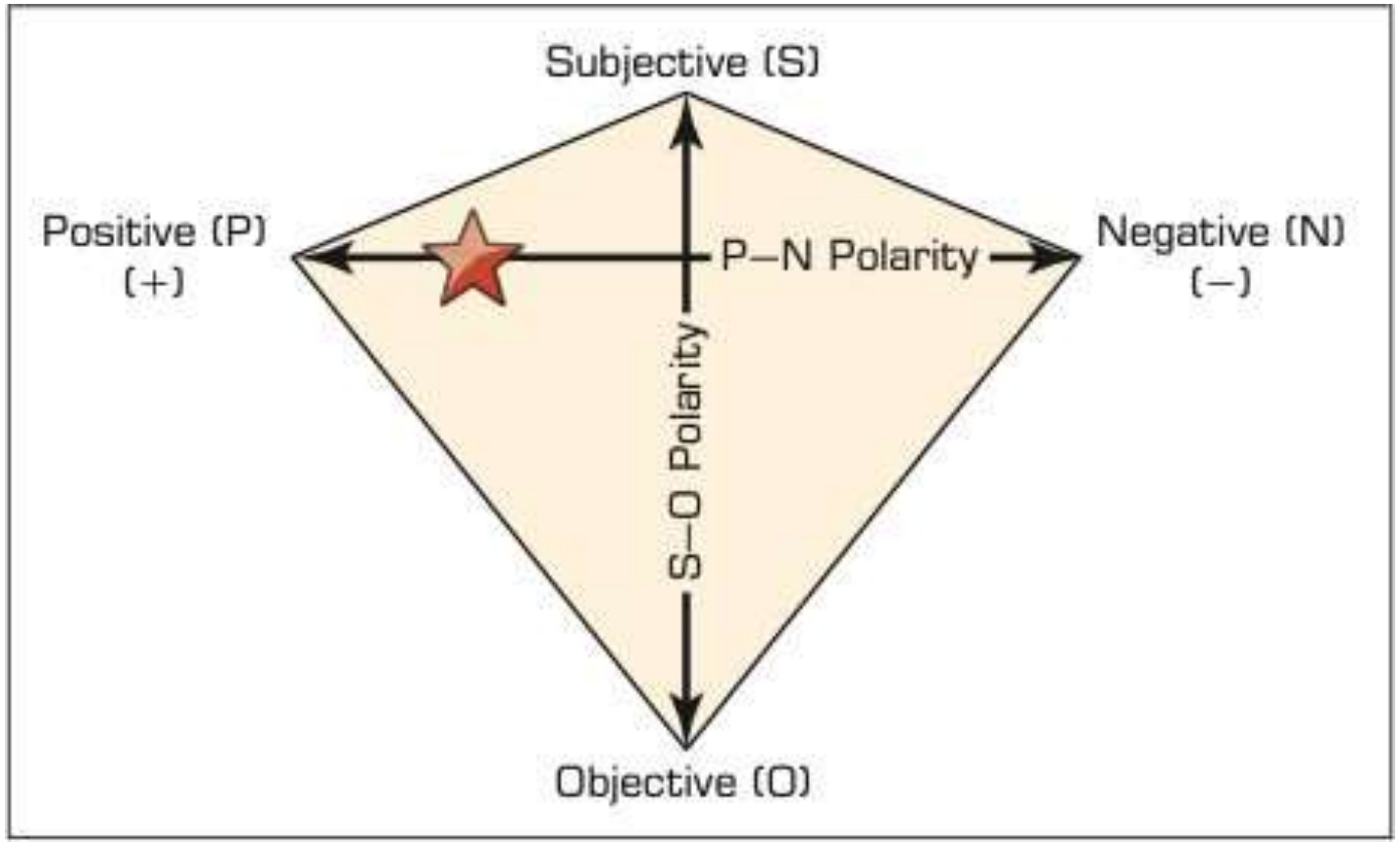
Polarity Identification – negative vs positive

- Can be made at the word, term, sentence, or document level.

- Two dominant techniques have been used for identification of polarity at word level:
  1. Using a lexicon as a reference library (e.g., WordNet, SentiWordNet)
  2. Using a collection of training documents
     - statistical analysis / machine learning
     - Large labeled textual data sets for prediction (e.g., Cornell Movie-Review, Stanford—Large Movie Review data sets).

- Identifying semantic orientation of sentences and phrases or documents
  - Extend semantic orientation of individual words to phrase, sentence or documents in which the word appears.
  - Aggregation typically accomplished by some type of sentiment polarity averaging.

P-N Polarity and S-O Polarity



P–N Polarity and S–O Polarity Relationship

7.7 Web Mining Overview

- Web Content and Web Structure Mining

7.8 Search Engines

- Anatomy of a Search Engine
- Search Engine Optimization
- Methods for Search Engine Optimization

7.8 Web Usage Mining (Web Analytics)

- Web Analytics Technologies
- Web Analytics Metrics

Web Mining Overview

- The growth of the Internet has made data creation, data collection, and data/information/opinion exchange easier.

- The amount of information on the Web is growing rapidly → creating the largest data and text repository.

- Web provides rich data for knowledge discovery (e.g., HTML ,XML, hyperlinks and usage information) → Web mining

- Due to the size and complexity, mining the Web is challenging
  - The Web is too big for effective data mining
  - The Web is too complex
  - The Web is too dynamic
  - The Web is not specific to a domain
  - The Web has everything

Web Mining Overview

- Web mining is the process of discovering intrinsic relationships (i.e., interesting and useful information) from Web data (textual, linkage, or usage information).

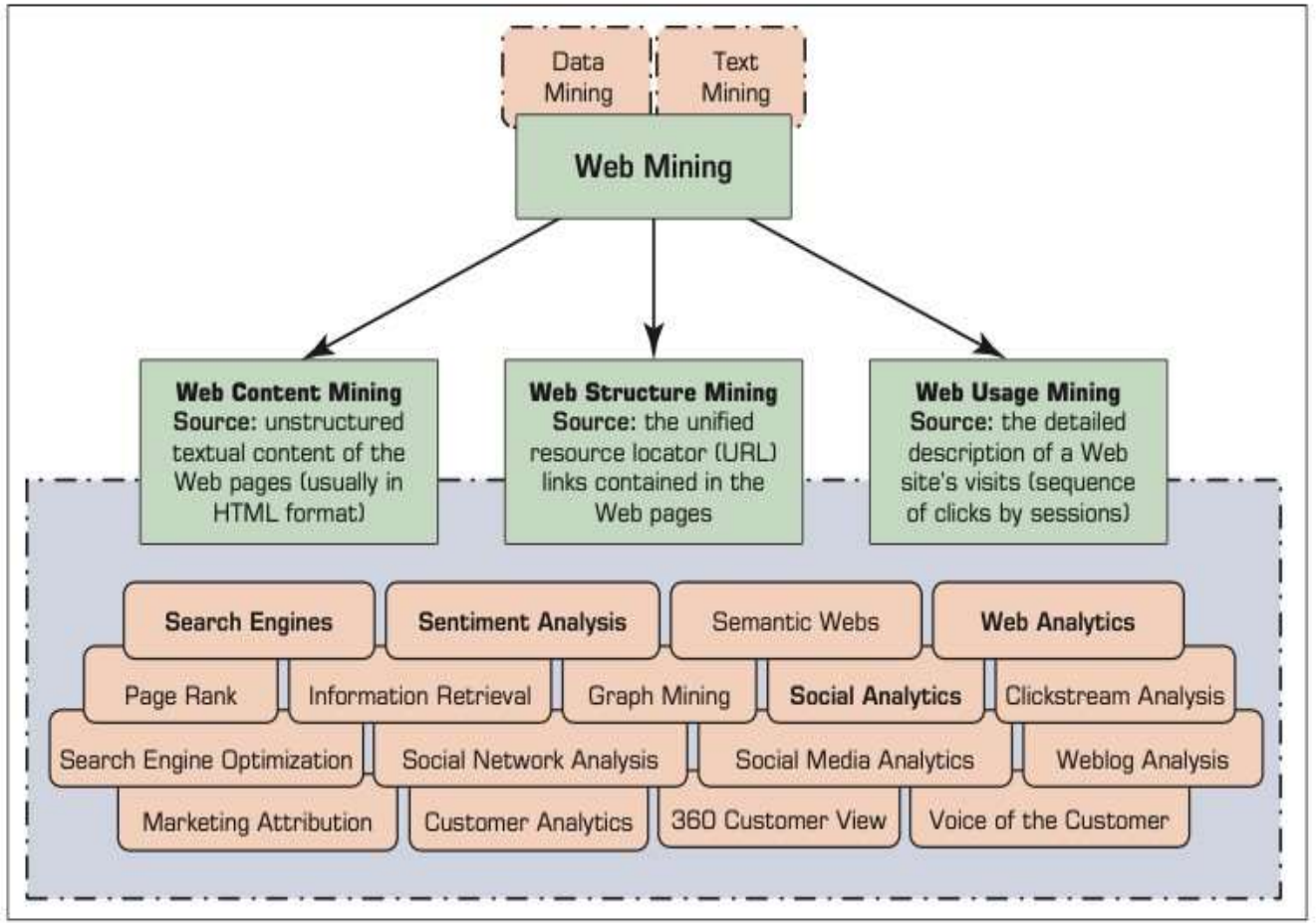
- The goal is to turn business transactions, customer interactions, and Web site usage data into actionable information (i.e., knowledge) to promote better decision making.

- It is the same as data mining but uses data generated over the Web.

Web Mining Overview

- Web mining consists of three areas:
  - Content mining
  - Structure mining
  - Usage mining
- Web Mining versus Web Analytics
  - Web analytics → Web site usage focused data
  - Web mining → all data generated via the Internet (i.e., transaction, social).

Taxonomy of Web Mining



Web Content and Web Structure Mining

- Automatic extraction of useful information from Web pages (textual content) → Web content mining

- Data are collected Web crawlers (a.k.a spiders).

- It can be used to enhance search results produced by search engines.

- Web pages include hyperlinks: Authoritative pages and Hubs
  - hyperlink-induced topic search (HITS): referenced algorithm used to calculate hubs and authorities

- Web structure mining generate interesting information from the links embedded in Web documents.

  - used to identify authoritative pages and hubs

Search Engines

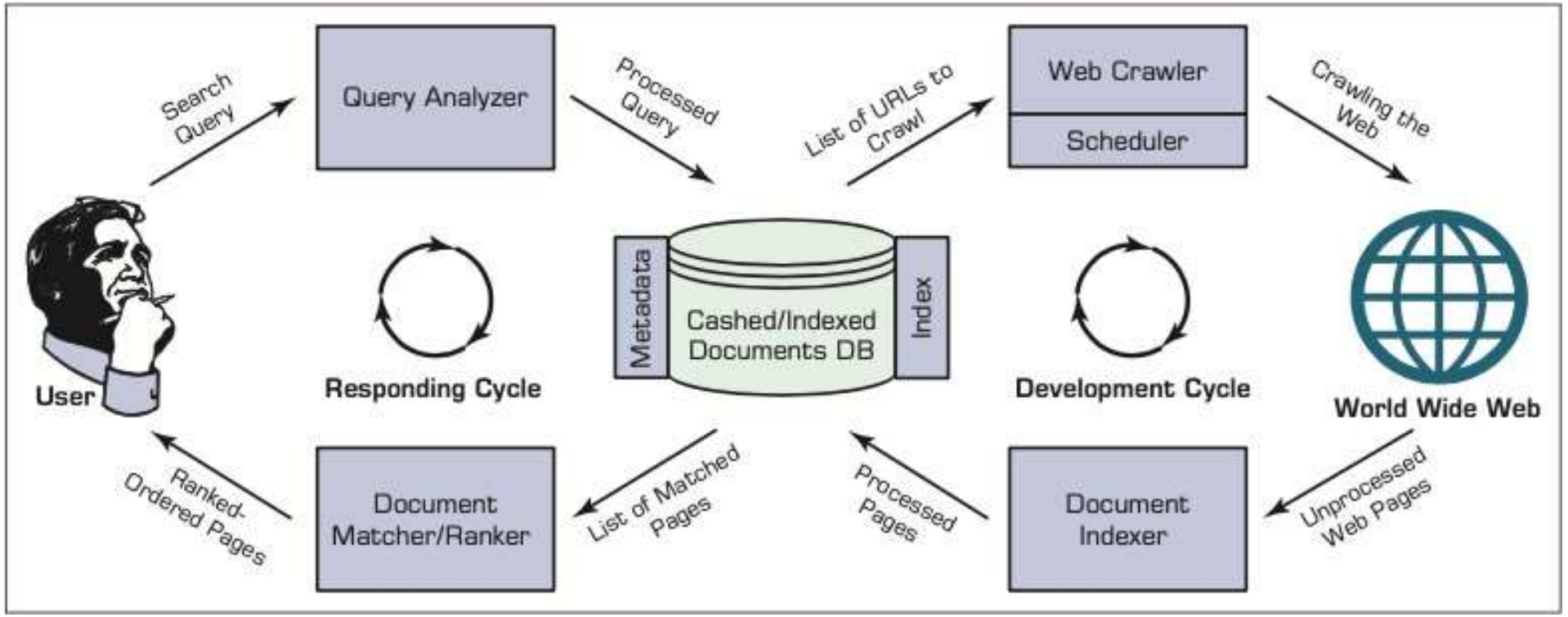
- Search engine→ Software program that searches for documents (Internet sites or files) based on the keywords (individual words, multi-word terms, or a complete sentence) users have provided that relate to the subject of their inquiry.
- E.g., Google, Bing, and Yahoo.

- Popular term for information retrieval system

- The two metrics to evaluate search engines:
  - Effectiveness/quality in finding the right documents/pages.
  - Efficiency/speed of returning a response quickly.

Search Engines

Structure of a Typical Internet Search Engine



Search Engine

Top 15 Most Popular Search Engines

Rank	Name	Estimated Unique Monthly Visitors
1	Google	1,600,000,000
2	Bing	400,000,000
3	Yahoo! Search	300,000,000
4	Ask	245,000,000
5	AOL Search	125,000,000
6	Wow	100,000,000
7	WebCrawler	65,000,000
8	MyWebSearch	60,000,000
9	Infospace	24,000,000
10	Info	13,500,000
11	DuckDuckGo	11,000,000
12	Contenko	10,500,000
13	Dogpile	7,500,000
14	Alhea	4,000,000
15	ixQuick	1,000,000

Anatomy of a Search Engine

- **Development Cycle** (two main components)

  - Web crawler: find and fetch web pages
  - Document indexer: process the web pages or files and place them into the document database.
  - Steps:
    1. Preprocessing The Documents
       - Collecting, formatting, and storing.
    2. Parsing The Documents
    3. Creating The Term-by-document Matrix
       - word/term values: numeric, binary, TF/IDF

Anatomy of a Search Engine

- **Response Cycle**
  - Query Analyzer
    - Responsible for receiving a search request from the and converting it into a standardized data structure.
  - Document Matcher/Ranker
    - The structured query data are matched against the document database to find the most relevant documents/pages and rank them in the order of relevance/importance.
    - Old approach - keyword match
    - Google approach - PageRank
      - Algorithmic way to rank order documents/pages based on their relevance and value/importance.

Search Engine Optimization

- SEO is the intentional activity of affecting the visibility of an e-commerce site or a Web site in a search engine's natural (unpaid or organic) search results.

- Part of Internet marketing strategy:
  - considers how search engines work
  - what people search for
  - the actual search terms or keywords typed into search engines
  - which search engines are preferred by their targeted audience

- Optimizing a Web site can involve editing its content, HTML, and associated coding to both increase its relevance to specific keywords and to remove barriers to the indexing activities of search engines.

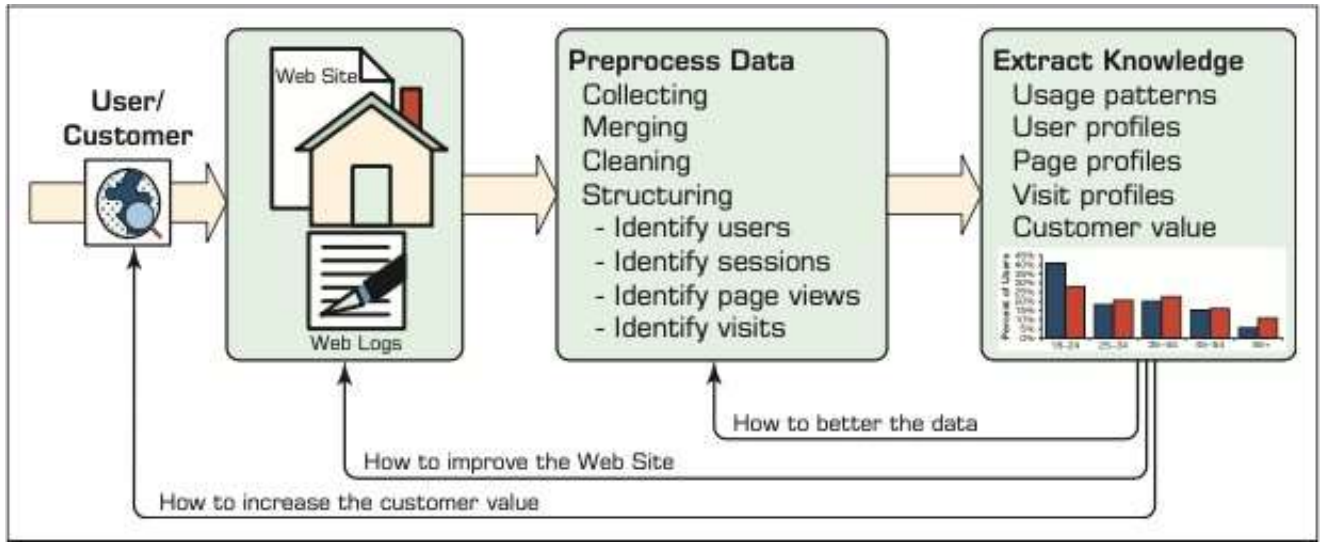
Methods for Search Engine Optimization

- **White-Hat SEO** search engines recommend technique
  - tend to produce results that based on good site design
  - conforms to the search engine's guidelines and involves no deception
  - ensuring that the content a search engine indexes and ranks is the same content a user will see.
- **Black-hat SEO** search engines do not approve
  - attempts to improve rankings in ways that are not approved by the search engines or involve deception.
  - spamdexing (a.k.a search spam, search engine spam, or search engine poisoning)

Web Usage Mining (Web Analytics)

Web usage mining is the extraction of useful information from data generated through Web page visits and transactions (e.g., server logs, cookies, page metadata).

- Analysis of the information collected by Web servers can help us better understand user behaviour → clickstream analysis
  - E.g., knowing when visitors access a site, determining where to place online advertisements.



Extraction of Knowledge from Web Usage Data

Web Analytics Technologies

- Web analytics tools help to measure, collect, and analyze Internet data to
  - better understand and optimize web usage and traffic.
  - to assess and improve the effectiveness of e-commerce Web sites.
  - help companies measure the results of traditional print or broadcast advertising campaigns

Two main categories of Web analytics:

- **Off-site Web analytics:** refers to Web measurement and analysis about you and your products that take place outside the Web site
  - E.g., Web site's potential audience, share of voice (visibility or word of mouth), and buzz (comments or opinions) that is happening on the Internet.
- **On-site Web analytics:** measure visitors' behavior once they are on the Web site.
  - E.g., drivers and conversions.
  - Google Analytics is the most widely used on-site Web analytics service.
  - to method are used to produce Web traffic reports: server log file analysis and page tagging.

Web Analytics Metrics

- Provide nearly real-time valuable marketing data.
- Metrics categories:
  - Web site usability: How were they using my Web site?
  - Traffic sources: Where did they come from?
  - Visitor profiles: What do my visitors look like?
  - Conversion statistics: What does it all mean for the business?

Web Analytics Metrics

Web Site Usability

1. Page views
2. Time on site
3. Downloads
4. Click map
5. Click paths

Visitor Profiles

1. Keywords
2. Content groupings
3. Geography
4. Time of day
5. Landing page

Traffic Sources

1. Referral Web sites
2. Search engines
3. Direct
4. Offline campaigns
5. Online campaigns

Conversion Statistics

1. New visitors
2. Returning visitors
3. Leads
4. Sales/conversions
5. Abandonment rates

7.10 Social Analytics

- Social Network Analysis
- Social Network Analysis Metrics
- Social media & analytics

Social Network Analysis

- Social network - social structure composed of individuals (or groups/organizations) linked to one another with some type of connections/relationships
  - Help study relationships between individuals, groups, organizations
- Social analytics- analyses of social structure and dynamics.
- Interdisciplinary field that emerged from social psychology, sociology, statistics, and graph theory.
- Typical social network types
  - Communication networks, community networks, criminal networks, and innovation networks.

Social Network Analysis Metrics

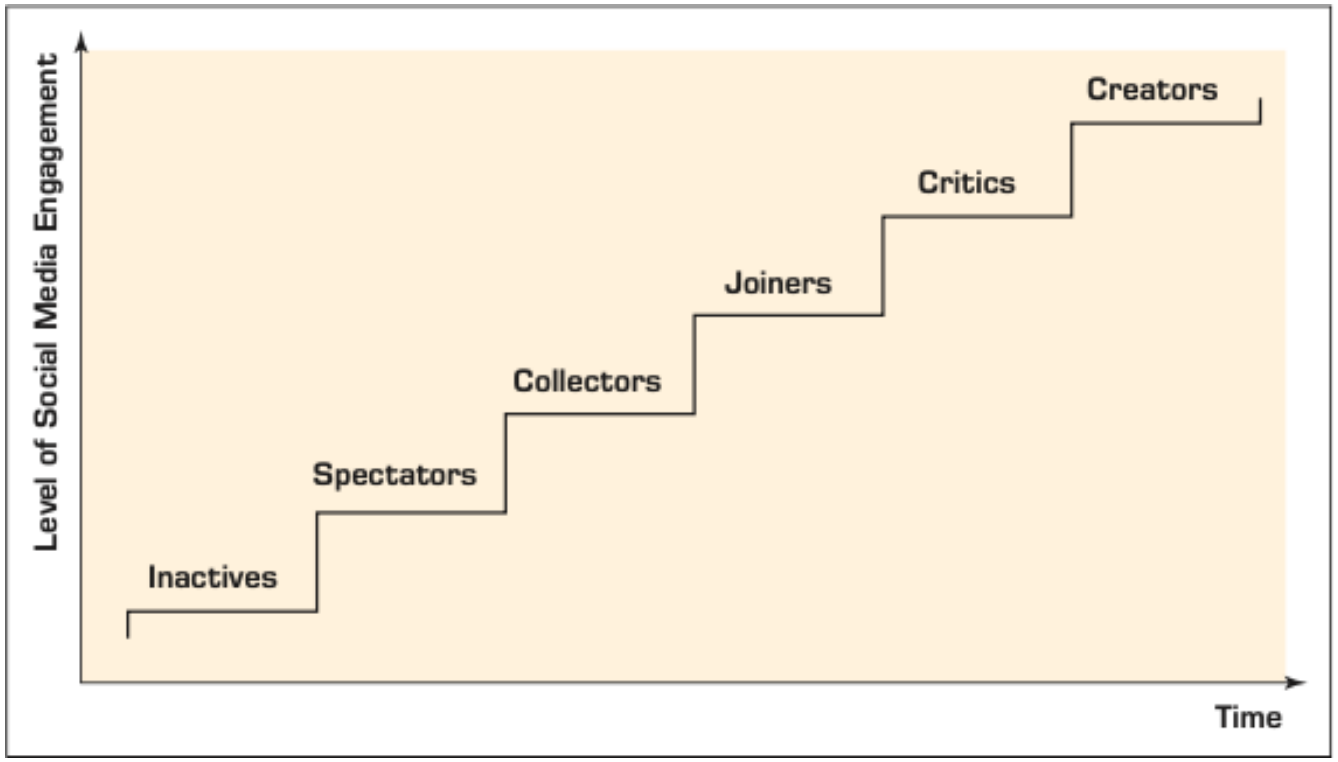
- Connections
  - Homophily
  - Multiplexity
  - Mutuality/reciprocity
  - Network closure
  - Propinquity
- Segmentation
  - Cliques and social circles
  - Clustering coefficient
  - Cohesion
- Distribution
  - Bridge
  - Centrality
  - Density
  - Structural holes
  - Tie strength

Social Media

- Enabling technologies of social interactions among people in virtual communities and networks (e.g., forums, social blogs, microblogging, wikis, social networks)
- Build on technological foundations of Web 2.0
- Social media are different from traditional/industrial media, such as newspapers, television, and film
  - Quality
  - Reach
  - Frequency
  - Accessibility
  - Usability
  - Immediacy
  - Updatability

Social Media

How Do People Use Social Media?



Six different engagement levels

Social media analytics

- Systematic and scientific ways to consume the vast amount of content created by Web-based social media outlets tools, and techniques for the betterment of an organization's competitiveness.

- Rapidly growing movement in analytics.

- Measuring the social media impact:
  - Descriptive analytics
  - Social network analysis
  - Advanced analytics

Social media analytics

Best Practices in Social Media Analytics

- Think of measurement as a guidance system, not a rating system
- Track the elusive sentiment
- Continuously improve the accuracy of text analysis
- Look at the ripple effect
- Look beyond the brand
- Identify your most powerful influencers
- Look closely at the accuracy of your analytic tool
- Incorporate social media intelligence into planning

Reference

- Ramesh Sharda, Dursun Delen, Efraim Turban; Analytics, Data Science, & Artificial Intelligence: Systems For Decision Support, 11ed, Global Edition, Pearson Education Limited, 2021, ISBN 10: 1 - 2 92-34155-6, ISBN 13: 978-1-292-34155-2

  Chapter 7: Text Mining, Sentiment Analysis, and Social Analytics

Recommended Reading

- Sentiment Analysis: A Definitive Guide. MonkeyLearn. https://monkeylearn.com/sentiment-analysis/

Recommended Video

- What is Text Mining? (2015, Oct 8). [Video]. YouTube. https://www.youtube.com/watch?v=I3cjbB38Z4A

- Natural Language Processing In 5 Minutes | What Is NLP And How Does It Work? | Simplilearn (2021, Mar 17). [Video]. YouTube. https://www.youtube.com/watch?v=CMrHM8a3hqw